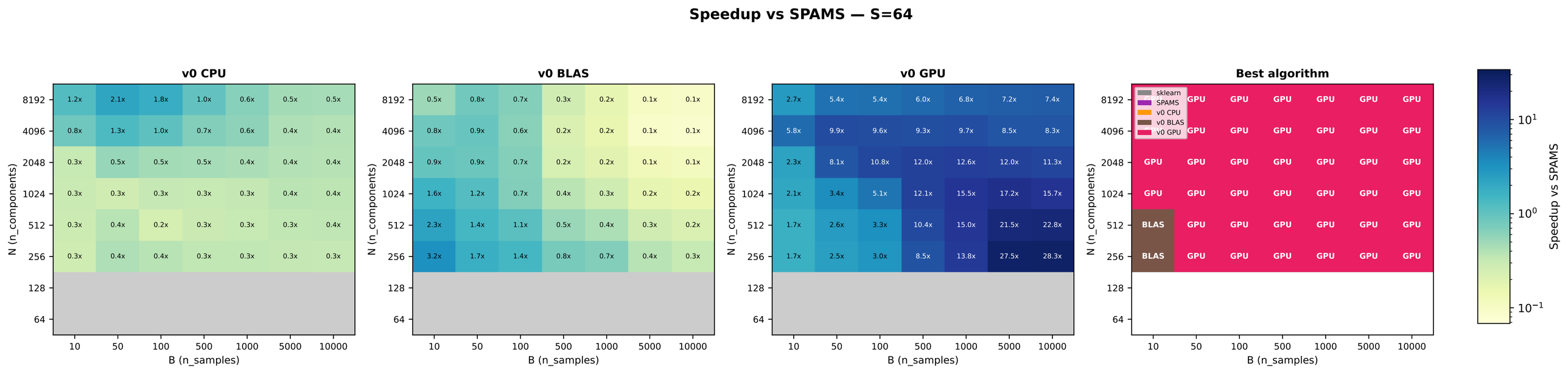
Speedup vs SPAMS — S=64
v0 CPU
v0 BLAS
v0 GPU
Best algorithm
N (n_components)
B (n_samples)
8192
4096
2048
1024
512
256
128
64
10
50
100
500
1000
5000
10000
1.2x 2.1x 1.8x 1.0x 0.6x 0.5x 0.5x
0.8x 1.3x 1.0x 0.7x 0.6x 0.4x 0.4x
0.3x 0.5x 0.5x 0.5x 0.4x 0.4x 0.4x
0.3x 0.3x 0.3x 0.3x 0.3x 0.4x 0.4x
0.3x 0.4x 0.2x 0.3x 0.3x 0.3x 0.4x
0.3x 0.4x 0.4x 0.3x 0.3x 0.3x 0.3x
0.5x 0.8x 0.7x 0.3x 0.2x 0.1x 0.1x
0.8x 0.9x 0.6x 0.2x 0.2x 0.1x 0.1x
0.9x 0.9x 0.7x 0.2x 0.2x 0.1x 0.1x
1.6x 1.2x 0.7x 0.4x 0.3x 0.2x 0.2x
2.3x 1.4x 1.1x 0.5x 0.4x 0.3x 0.2x
3.2x 1.7x 1.4x 0.8x 0.7x 0.4x 0.3x
2.7x 5.4x 5.4x 6.0x 6.8x 7.2x 7.4x
5.8x 9.9x 9.6x 9.3x 9.7x 8.5x 8.3x
2.3x 8.1x 10.8x 12.0x 12.6x 12.0x 11.3x
2.1x 3.4x 5.1x 12.1x 15.5x 17.2x 15.7x
1.7x 2.6x 3.3x 10.4x 15.0x 21.5x 22.8x
1.7x 2.5x 3.0x 8.5x 13.8x 27.5x 28.3x
sklearn
SPAMS
v0 CPU
v0 BLAS
v0 GPU
BLAS
GPU
Speedup vs SPAMS
10^{1}
10^{0}
10^{-1}